ENGINEERING
PORTFOLIO / 2026

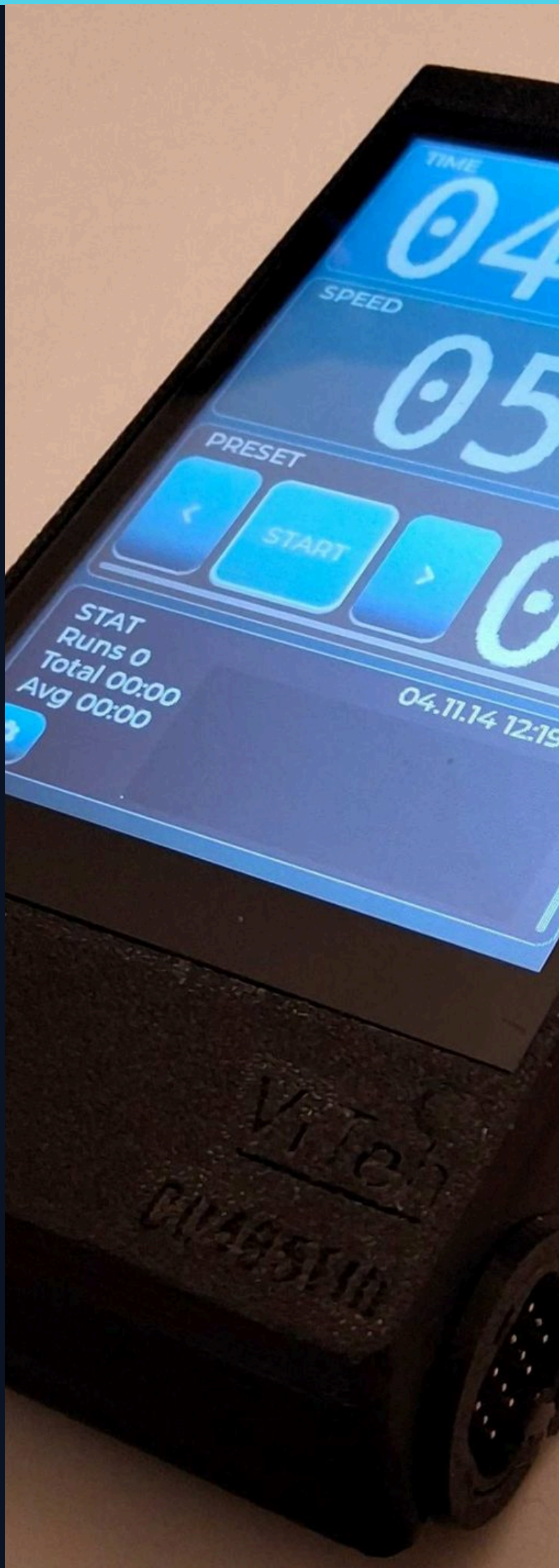
VITALII KRUTIKOV

Embedded Systems Engineer
Hardware & PCB Designer
Software Developer

From circuit design and firmware to production-oriented prototypes, desktop tools, websites, Android apps and connected systems.

STM32 · ESP32 · C/C++ · C#
LINUX · PCB · IoT · AI

SELECTED WORK · VISIBLE PART OF WORK SINCE 2014



01 / PROFILE

Engineering, end to end

Product development across embedded electronics, desktop software, web, Android, automation, networking and prototyping.

PROFESSIONAL PROFILE

Engineer and developer working since 2014 (12+ years), spanning embedded systems, PCB design, automation and PC software. Work includes sensor products, industrial controllers, telemetry devices, motor-control electronics and engineering utilities. Recent work adds websites and Android applications for private AI systems.

Experience: IT engineer at Adventx and Tesla-service, plus freelance projects.

CORE CAPABILITIES

01 EMBEDDED	STM32 (F0–F7, G0/G3/G4), ESP32, AVR; C/C++; peripherals, drivers and integration.
02 ELECTRONICS	Schematics, multilayer PCBs, prototyping, design reviews and enclosure CAD.
03 CONNECTIVITY	CAN, Modbus RTU, RS-485, UART, USB, SPI, I²C, Ethernet, TCP/UDP, BLE and GSM.
04 SOFTWARE	Desktop apps: C# (WinForms, WPF), Qt/C++, GTK, Swift; Python, Bash, diagnostics and automation.
05 WEB & MOBILE	Fast static websites, web interfaces and admin panels (HTML/CSS/JS, Cloudflare); Android apps in Kotlin and Jetpack Compose.
06 DELIVERY	Device configuration, troubleshooting, testing tools, technical documentation and visual instructions.

02 / FEATURED PROTOTYPE

Touchscreen control unit

Recent physical prototype photographed in November 2026.

CUSTOM DEVICE

A compact, enclosed control system with an integrated color touchscreen, large numerical readouts, preset selection, start control and run statistics.

The photographed interface displays time, speed, preset navigation and status. The enclosure appears custom-made; its internal circuitry and firmware specifications are not established by the available photograph.

DESIGN FOCUS Human-machine interface · Enclosure integration · Device-level controls



Touchscreen control unit, prototype photo

02A / 3D MODELS

Controller in 3D

Two generations of the touchscreen control unit as web-ready 3D models (GLB), exported from the original Blender project; the renders below come from those GLB files.



First version: small display, buttons, black cover



Latest version: large touchscreen, no buttons, ViTeh mark, round connector

MODELS FOR THE WEB

Lightweight GLB models (about 150 KB each) with PBR materials, the screen texture taken from a photo of the real unit; they plug into the interactive 3D product gallery on viteh.org.

SKILLS · Blender / glTF / WebGL

FROM CAD TO PRODUCT PAGE

The path from the enclosure design and PCB project (KiCad, Blender) to an interactive model that visitors can rotate, without third-party viewers or tracking.

SKILLS · enclosure CAD / 3D pipeline / web

03 / INDUSTRIAL CONTROL

Sensors & automation

Selected projects documented with original product and system images.



Greenhouse screen controller

GREENHOUSE CONTROLLER

Controller for opening and closing greenhouse screens, with a local display and associated control electronics. The original project archive includes connection illustrations and a video guide.



DMM-T2 sensor module

DMM-T2 SENSOR MODULE

Temperature and humidity data acquisition modules with Modbus RTU communication, including variants for thermocouple inputs and different mechanical layouts.



Multi-channel temperature display



Ethernet-connected temperature monitor

04 / ELECTRONICS & PCB

Custom hardware platforms

Original board imagery from previous hardware development projects.



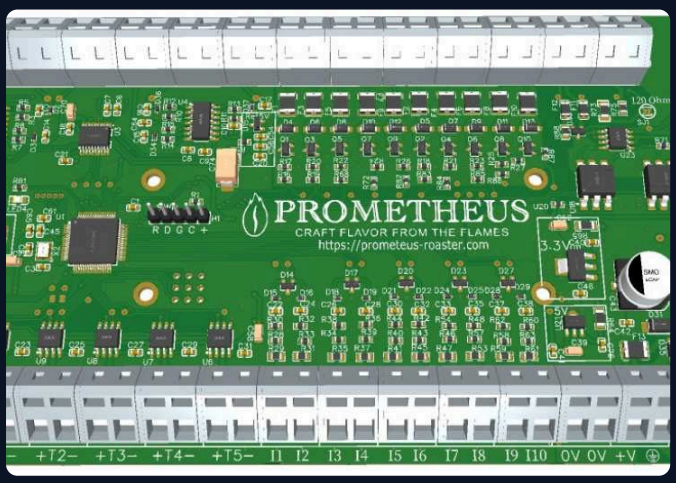
TryBrd / multi-layer stepper-control PCB

MULTI-LAYER MOTION CONTROL

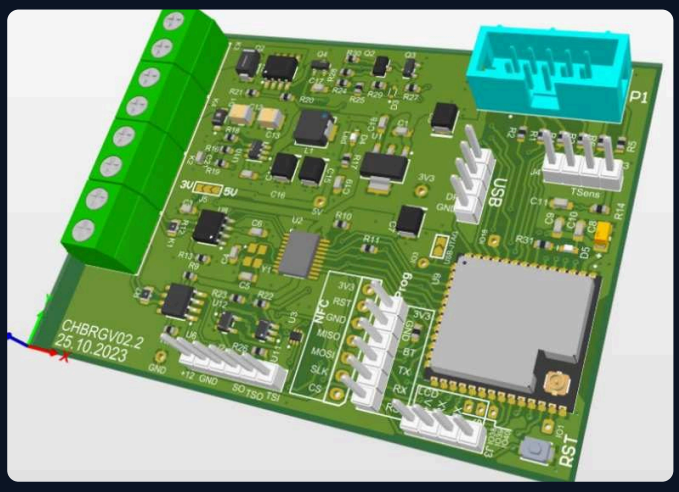
Designed a four-layer controller board for TMC5130A stepper motor drivers as part of a customer 3D-printing project; also developed basic motor-control firmware. Additional expansion-board work addressed multi-motor systems.

OTHER BOARD WORK

I/O electronics, charging-station modules, GSM monitoring, and current ESP32-S3 gateway / Ethernet / RS-485 / CAN PCB work.



Custom I/O expansion board

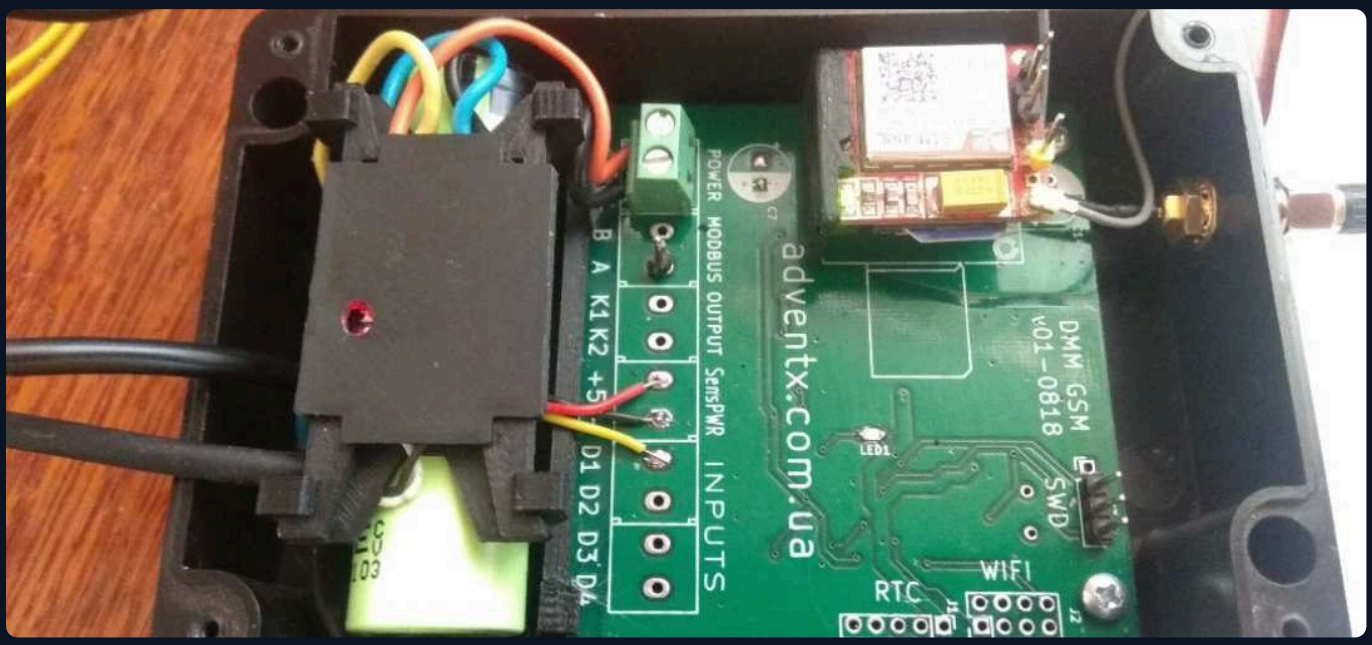


CHBRG charging-station device board

05 / CONNECTED DEVICES

Telemetry & remote monitoring

Field electronics designed to collect information, report status and integrate with operator systems.



GSM event and sensor logger hardware

GSM SENSOR LOGGER

Four sensor inputs, two relay outputs and backup power. The logger recorded sensor values and issued SMS alerts when configured thresholds were exceeded. It also uploaded data to a server for charting and remote configuration, with SMS-based commands.

ADDITIONAL CONNECTED PRODUCTS

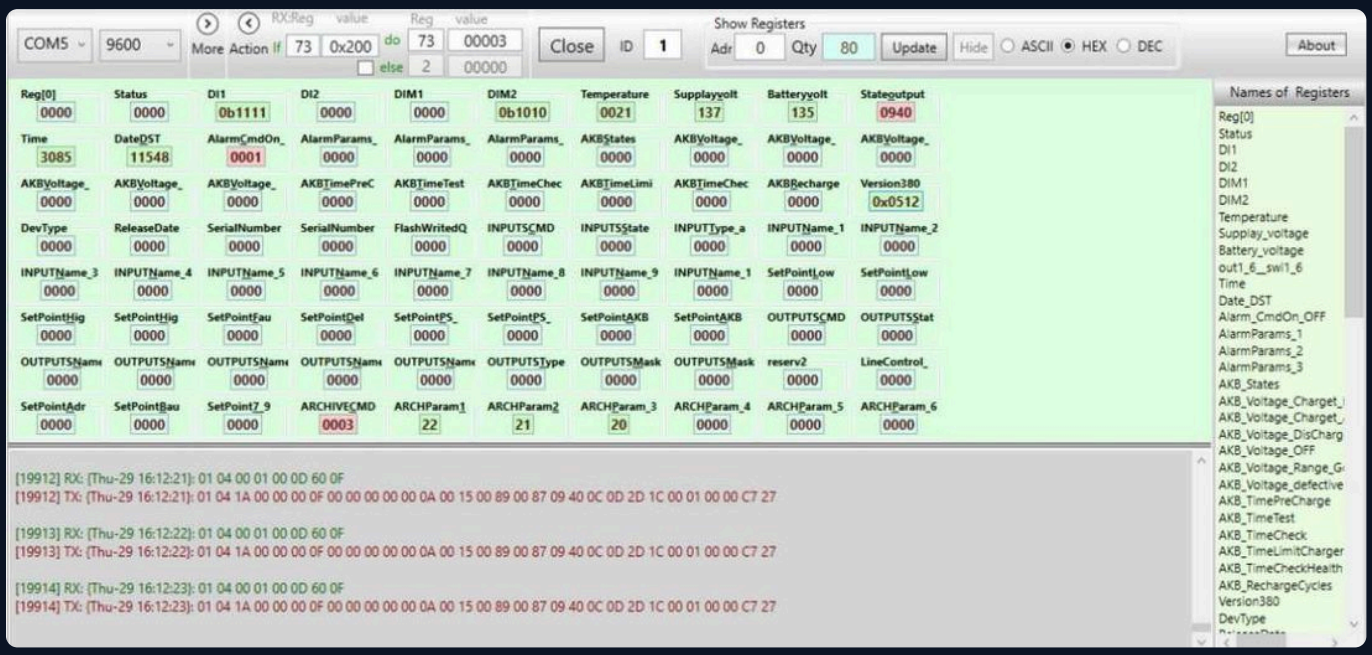
Broadcast watchdog: GSM-based radio-audio monitoring with loss-of-broadcast alerts and remote audio listening.

Event logger: PIR event timestamps collected to a PC utility, tabular viewer and CSV export; enclosure produced with 3D printing.

06 / SOFTWARE ENGINEERING

Tools that support hardware

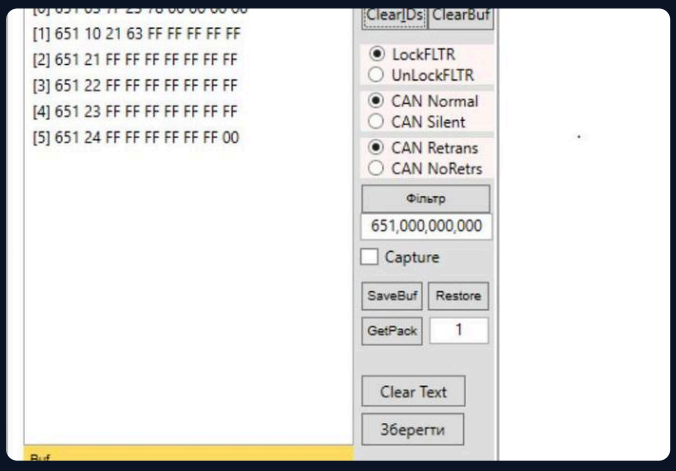
Desktop control applications, diagnostic utilities and cross-platform engineering workflows.



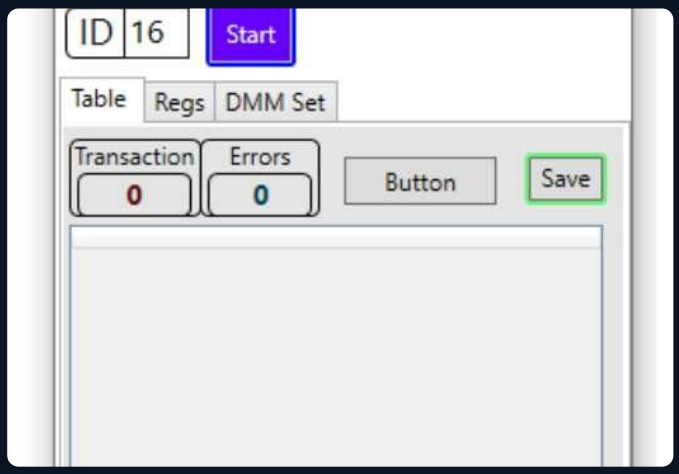
Modbus RTU diagnostic and register-monitoring utility

DESKTOP TOOLS

C# software for sensor configuration and control, serial-port communication, Modbus analysis, CAN packet inspection and device diagnostics. Includes custom user interfaces for engineering and service tasks.



CAN analysis / packet utility

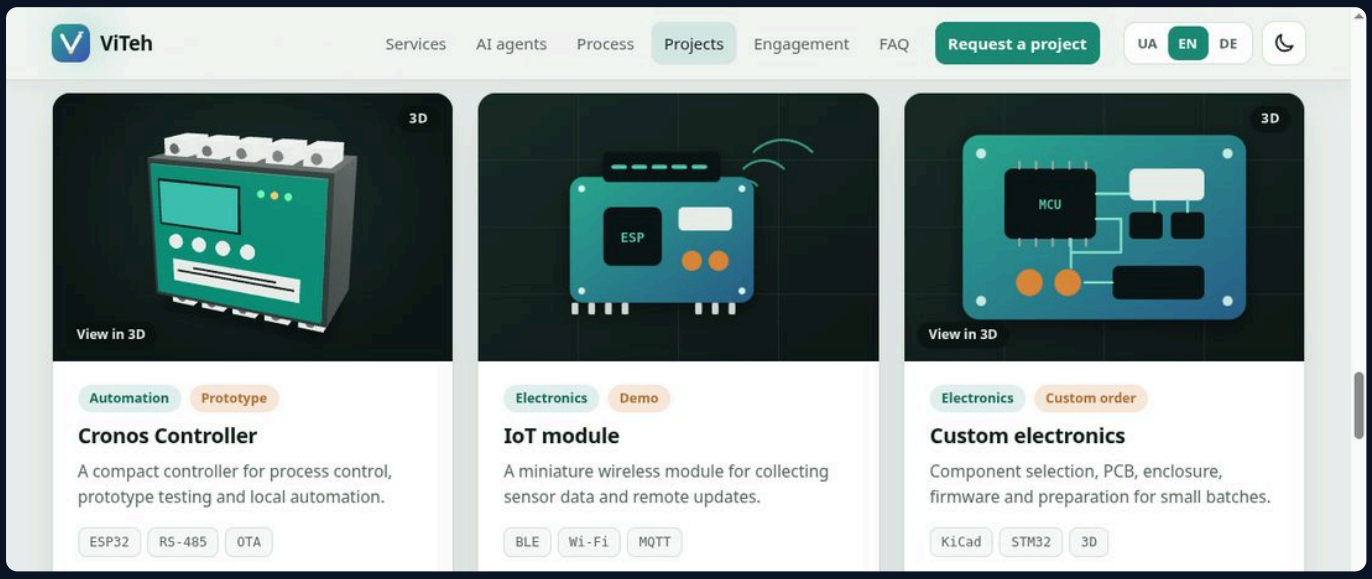


DMM configuration application

06A / WEB DEVELOPMENT

Websites & web interfaces

Fast, private, multilingual sites, with the tools to run them.



viteh.org: product cards with a 3D view (UA / EN / DE, light and dark theme)

STATIC SITES, NO TRACKERS

Static, no-build sites without third-party requests (no CDN, fonts or analytics), built to be GDPR-friendly, with legal pages for Germany and light/dark themes.

SKILLS · web design / performance / privacy

CONTACT FORMS & INTEGRATIONS

Contact form with attachments served by a Cloudflare Worker and Email Routing, with spam protection and origin checks; hooks for mail and chat assistants.

SKILLS · Cloudflare Workers / forms / integrations

3D PRODUCT GALLERY

Interactive 3D models of hardware products (GLB exported from Blender), loaded lazily, one scene at a time, with a flat-image fallback.

SKILLS · 3D on the web / Blender / accessibility

PRIVATE ADMIN & TEMPLATES

A private admin for the site: switch sections on or off, edit texts in three languages, build pages from templates, preview, then publish.

SKILLS · admin tooling / content workflow

06B / ANDROID APPLICATIONS

Android client for private AI systems

Native Android application for the user's own servers and devices.

CHAT & VOICE AGENT

Kotlin and Jetpack Compose app with a chat tab and a voice-agent tab: speech recognition, spoken replies, headset button controls and foreground services.

SKILLS · Kotlin / Compose / audio pipelines

INTERPRETER & LECTURE MODE

Live translation and lecture capture with speech segmentation and voice detection, running against local speech and language models.

SKILLS · speech processing / streaming / UX

FILES, MAIL & AGENT BOARD

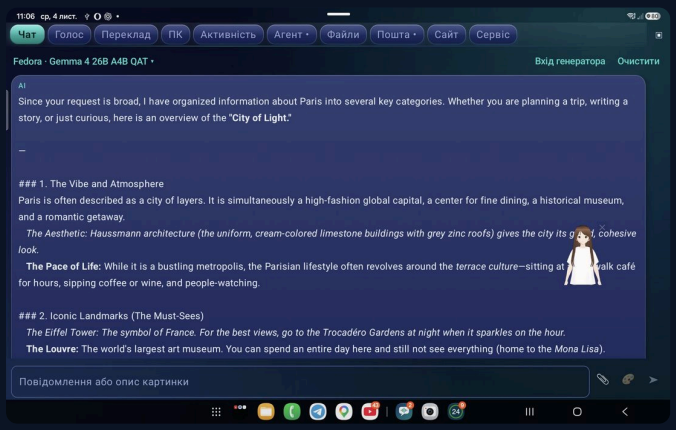
Files tab with previews, a mail tab with summaries and push-style notifications, and a board for agent tasks.

SKILLS · app architecture / notifications / protocols

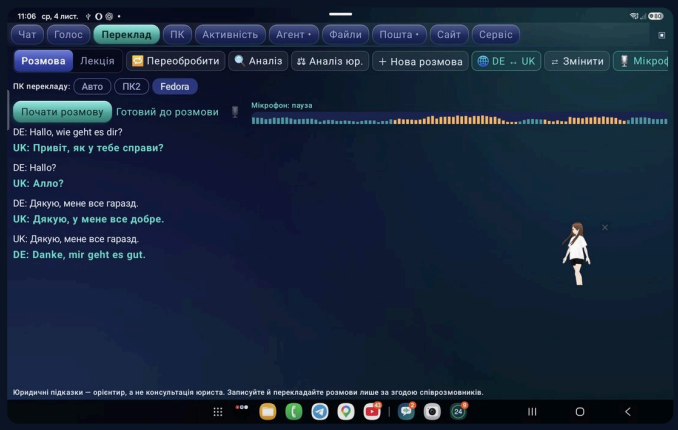
PRIVATE-NETWORK OPERATIONS

Routing over a private tailnet with fallbacks, automatic silent updates, on-device logs that can be uploaded for diagnostics, and a scripted release process.

SKILLS · networking / delivery / maintenance



Chat

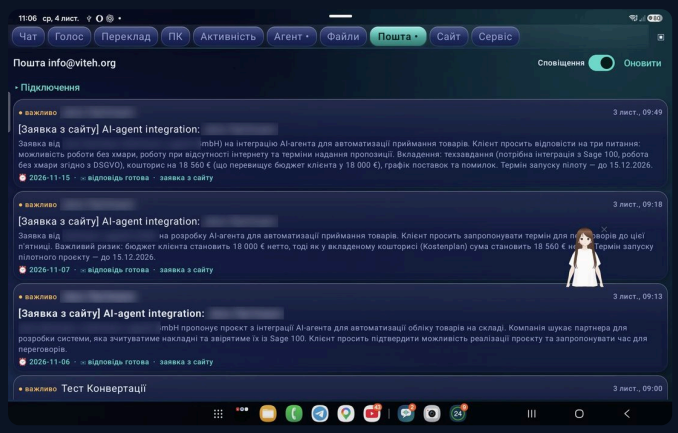


Live translation

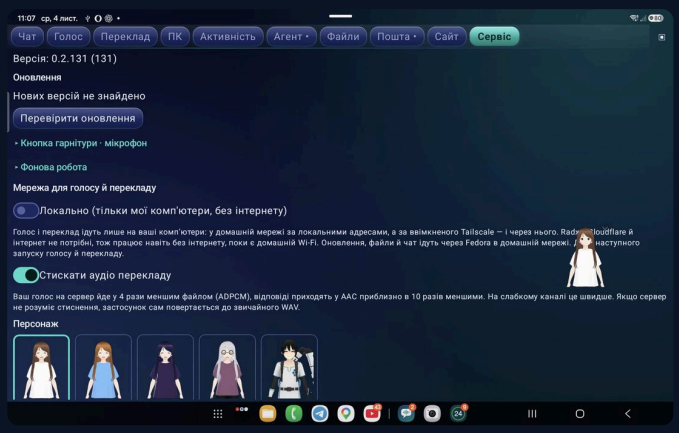
06C / AI CLIENT IN USE

Private AI client: web and Android

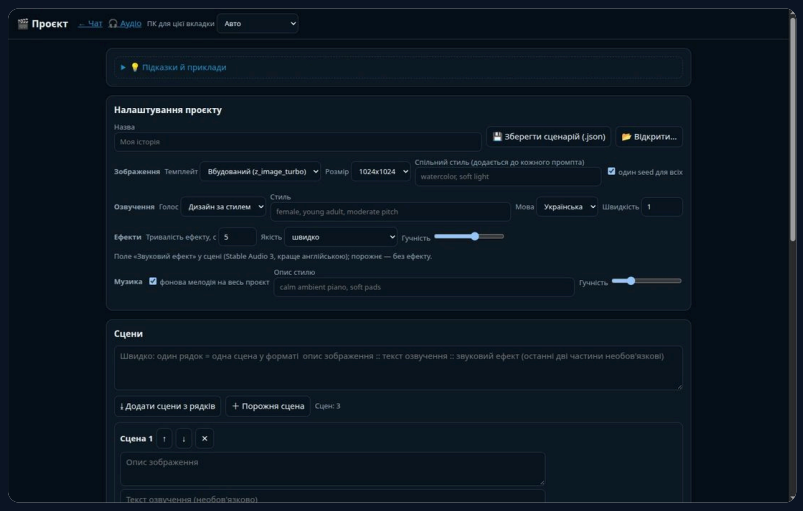
The same private AI system reached from a browser and from a native tablet app (screens with test data; private texts blurred).



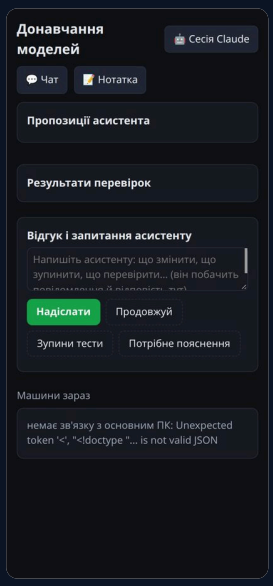
Mail agent (test messages)



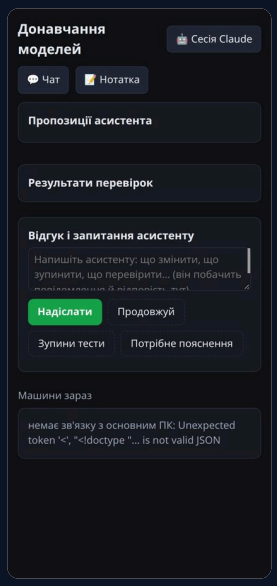
Service status



Web: project page, light theme



Web on a phone, dark theme



Web on a phone, light theme

07 / SERVICES

What I offer

The same service names as on viteh.org, each backed by the work shown in this portfolio.

Software development

Web apps, internal tools, APIs and desktop utilities tailored to your workflows.

Web and cloud · Integrations with your systems · Documentation and code handover

Electronic devices

From schematic and PCB to firmware and enclosure — prototypes and small series.

Schematics and PCB layout · Firmware (ESP32, STM32, RP2040) · Enclosures and 3D printing

AI-agent integration

Agents that read documents, talk to your systems and handle routine work under your control.

Connected to your data (RAG) · Tool and API calling · Cloud or local models

IoT and automation

Sensors, gateways and dashboards: turning device data into a clear picture.

MQTT, Wi-Fi, BLE, LoRa · OTA firmware updates · Dashboards and alerts

Prototypes and proof of concept

We test your idea quickly on a real device or demo before you invest more.

Working prototype within weeks · Clear success criteria · Scale-up recommendations

Technical audit and consulting

An independent view on architecture, component choice, security and code quality.

Architecture review · Risk and cost assessment · Security by default

Website development

Fast multilingual websites without trackers or third-party requests, forms with attachments, private admin tools and 3D product galleries.

Static sites without a build step · Forms, mail, Cloudflare Workers · 3D product models and a private admin

Android apps

Native apps in Kotlin and Jetpack Compose for your own servers and devices: chat, voice, files, notifications.

Kotlin and Jetpack Compose · Voice, translation, background services · Automatic updates and private networks

Custom apps built together with AI agents

Apps for almost any platform (Windows, Linux, macOS/iOS, Android, web), developed together with AI agents, with tests and support.

Prior experience: C# (WinForms, WPF), Qt, GTK, Swift · Current examples: Android client, Rust/Slint, web tools · Tests and support

Local AI and voice

Running local language models on your own hardware, voice interfaces and fine-tuning.

Local LLMs, quantisation (GGUF) · Speech recognition and synthesis · Model fine-tuning

Linux, networks and diagnostics

Linux/OpenWrt setup, remote access, and diagnosis of hangs, memory and network problems.

Linux, OpenWrt, systemd · Tailscale, Cloudflare Tunnel · Log analysis and debugging

Desktop and cross-platform apps (C# WinForms/WPF, Qt, GTK, Swift): earlier projects on these platforms were not preserved and are not shown as case studies. They are offered as custom development together with AI agents; current examples are the Android client, a Rust/Slint application, web tools and a voice assistant.

08 / EMBEDDED R&D

Embedded Systems & HMI

Research, experiments and prototypes are part of my applied engineering expertise, not just future product ideas.

ESP32-S3 / LVGL GRAPHICAL INTERFACES

Implementation and evaluation of LVGL 9.x touchscreen user interfaces; display bring-up using CO5300, SH8601, NV3041A and ST7789 controllers; SPI/QSPI display integration, FT3168 touch and PSRAM constraints.

SKILLS · UI architecture / display drivers / HMI debugging

POWER MANAGEMENT & WIRELESS CONTROL

ESP32-C6 + ESP-NOW servo-control experiments with wake, transmission and sleep cycles. Evaluated deep-sleep and light-sleep behavior, power consumption, PWM control and responsiveness.

SKILLS · low-power design / telemetry / measurement

MCU PLATFORMS & BOARD BRING-UP

Experience exploring STM32 and ESP32 alongside RISC-V and Linux-capable boards including Milk-V Duo, LicheeRV Nano and RK3506-based hardware. Evaluated peripheral capabilities and development workflows.

SKILLS · platform evaluation / firmware integration

EDGE HARDWARE & PCB ARCHITECTURE

Design investigation of connected microcontroller boards with CAN, RS-485, Ethernet, digital I/O and sensor interfaces; experience considering board layout, interoperability and physical integration.

SKILLS · schematics / interface selection / prototyping

09 / SYSTEMS R&D

Linux, Networks & Diagnostics

Hands-on system integration, debugging, protocol analysis and real-device experimentation.

LINUX / OPENWRT NETWORKING

Configured and investigated OpenWrt gateways, virtualized routers, IPv4 routing, LAN bridges, Tailscale subnet routing, Cloudflare Tunnel and remote management workflows.

SKILLS · networking / firewall logic / remote access

SINGLE-BOARD COMPUTERS & SERVICES

Raspberry Pi and embedded Linux platform investigations: storage and boot planning, SSH, services, Docker-based workflows and device-oriented Linux automation.

SKILLS · Linux administration / Bash / systems integration

AUTOMOTIVE CAN BUS & DIAGNOSTICS

Practical Toyota Prius hybrid-battery repair and diagnostic work with CANable V2, Python-CAN, ECU request/response traffic, battery-block voltage and regeneration/load measurements.

SKILLS · CAN / diagnostics / data interpretation / hardware repair

DESKTOP & DEVICE INTEGRATION

Experience solving hardware/software integration problems across Linux and Windows, including device interfaces, remote desktop, printing services, USB, serial tools and firmware processes.

SKILLS · troubleshooting / integration / automation

10 / COMPUTE R&D

Local AI & GPU Engineering

Applied investigations that developed demonstrable technical knowledge in model deployment and heterogeneous computing.

LOCAL LLM DEPLOYMENT

Hands-on operation and testing of coding and general-purpose local models, including GGUF quantized deployments. Compared model quality, quantization, memory footprint and latency trade-offs.

SKILLS · inference deployment / benchmarking / evaluation

AMD ROCm / MEMORY MANAGEMENT

Troubleshot ROCm model-loading behavior on an AMD unified-memory system, including mmap versus mlock loading, memory limits and model residency. Documented successful tool-call tests in personal experiments.

SKILLS · ROCm / runtime debugging / memory profiling

MULTI-GPU / EXPERT PLACEMENT

Explored NVIDIA and AMD compute configurations, GPU VRAM limits, CPU/GPU offload, MoE active-expert placement, partitioning and the potential impact on inference throughput.

SKILLS · hardware-aware optimization / performance analysis

AI-ASSISTED ENGINEERING WORKFLOWS

Experimented with local coding assistants, tool-calling reliability, scripting, Git-based workflows and repeatable technical checks for embedded and software development.

SKILLS · developer tooling / validation / agent workflows

11 / COLLABORATION

Building real systems.

I work across the boundary of hardware and software: from embedded interfaces, sensing and communication to PC utilities, websites, Android applications, troubleshooting tools and physical product prototypes.

RELEVANT ROLES

- Embedded Systems Engineer / Firmware Developer
- Electronics & PCB Designer / IoT & Automation Developer
- Hardware–Software Integration Engineer
- Web & Android Application Developer

TECHNOLOGY

C · C++ · C# (WinForms, WPF) · Qt · GTK · Swift · Python · Bash · Kotlin · Jetpack Compose · Rust / Slint · HTML/CSS/JS · Cloudflare · STM32 · ESP32 · Linux · KiCad · Altium · Modbus · CAN · LVGL

CONTACT

info@viteh.org · viteh.org

This portfolio documents the visible part of the work done since 2014; there were more projects, including internal and tooling work that is not shown. It was first compiled in 2023 from project archives and original photographs, and updated in 2026 with recent work. Historical products, current prototypes and ongoing R&D are identified separately. Images are from the supplied project archive unless otherwise noted.